

CLEMENT TOH

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EDUCATION

National University of Singapore (NUS)

- Bachelor of Engineering (Mechanical Engineering) with Honours
- Relevant Coursework: Thermodynamics, Circuit Analysis, Additive Manufacturing

Singapore
Expected May 2028
GPA: 4.19 / 5.00

Singapore Polytechnic (SP)

- Diploma in Mechanical Engineering (with Merit)

April 2023
GPA: 3.86 / 4.00

TECHNICAL SKILLS

- CAD & Manufacturing:** SolidWorks, Siemens NX, Autodesk Inventor, Fusion 360, DFM, BOM, Additive Manufacturing (LPBF & FDM)
- Data & Software:** Python (Test Automation, Data Pipelines), C++, InfluxDB, Grafana, MATLAB, PlatformIO, GitLab
- Systems Integration & Embedded:** M5Stack/ IIoT Edge Devices, Arduino, PLCs, EPL (Ethernet POWERLINK) Network Analysis, Electro-Mechanical Troubleshooting, Hardware – Software Integration
- Test, Validation & Instrumentation:** EVT, Root Cause Analysis, NI DAQ (NI-9236), Oscilloscopes, Strain Gauges, Acoustic Emission Sensors, Optical QC, Class 4 Laser Safety & QC

PROFESSIONAL EXPERIENCE

JD Union Pte. Ltd.

Mechatronics Engineering Intern

Singapore

June 2026 – August 2026

- Root Cause Failure Analysis:** Executed Engineering Validation Testing (EVT) and optical microscopy on Class 4 industrial lasers (LUXINAR CO2), inspecting Quartz Block Head (QBH) fiber optics to prevent catastrophic window failures and eliminate optic downtime.
- Tooling & Fixture Design:** Designed a >250kg 5-tier mobile test workstation in SolidWorks using 4040 aluminium extrusion, dampening mechanical vibrations during sub-micron laser, ensuring DFM compliance for CNC fabrication.
- High-Precision Assembly & Integration:** Commissioned and integrated laser subsystems (IPG fiber laser, chiller, power supply), routing cable and component layouts to prevent thermal degradation within an enclosed chassis.

Advanced Remanufacturing and Technology Centre (A*STAR)

Systems Integration & Development Engineering Intern

Singapore

September 2022 – April 2023

- Preserved Chamber Integrity:** Engineered custom 13mm feedthrough ports and 3D-printed PCB holders (Siemens NX) to route 12 sensor channels into an EOS M 290 printer, maintaining a sealed inert-gas atmosphere.
- Eliminated Ignition Risks:** Integrated a 200mA fast-acting fuse into the active sensor signal path, eliminating electrical spark risks in a reactive metal powder environment.
- Isolate Micro-Deviations in Real-Time:** Built an end-to-end IIoT telemetry pipeline (12-ch NI DAQ, M5Stack, C++, InfluxDB, Grafana) to aggregate multi-parameter process metrics, successfully isolating powder-bed surface defects.
- Boosted Efficiency by 50%:** Developed Python scripts to tap and decode Ethernet POWERLINK traffic over a 30.6-hour print, implementing data-reduction algorithms cutting 50% overhead, retaining resolution to trace recoater faults.

Singapore Armed Forces (Armour Simulation Wing)

Systems Operator (National Service)

Singapore

May 2023 – May 2025

- Hardware System Uptime:** Maintained and debugged networked military simulation hardware alongside defense firms, resolving system faults to achieve operational stability across synchronized training cabins.

TECHNICAL PROJECT

3D Printer Thermal & Hardware Optimization (Ender 3 V3 SE)

June 2024

- Tooling & Airflow Optimization:** Redesigned print-head shroud in Fusion 360 and integrated high-performance dual blowers, optimizing airflow to resolve thermal inconsistencies and eliminate geometric distortion on steep overhangs.

LEADERSHIP & COMMUNITY INVOLVEMENT

Boon Lay Youth Network

Grassroots Leader

Singapore

January 2022 – December 2023

- Scaled Tutoring Operations:** Directed online academic support programs for underprivileged youth, managing 20 volunteer educators across 20 concurrent breakout sessions to maintain seamless tutoring delivery.